

1. (20 points)

Given:

- $X \sim \text{Uniform}[0, 10]$
- $Y \sim \text{Uniform}[2, 8]$
- X and Y are independent
- $Z = X + Y$

1. Calculate $\mu_X = E[X]$:

For a uniform distribution on $[a, b]$, the mean is:

$$E[X] = \frac{a+b}{2} = \frac{0+10}{2} = 5$$

Thus:

$$\boxed{\mu_X = 5}$$

2. Calculate $\mu_Z = E[Z]$:

Since $Z = X + Y$ and X, Y are independent:

$$E[Z] = E[X + Y] = E[X] + E[Y]$$

We already have $E[X] = 5$, and:

$$E[Y] = \frac{2+8}{2} = 5$$

Thus:

$$E[Z] = 5 + 5 = 10$$

$$\boxed{\mu_Z = 10}$$

3. Calculate $\sigma_X^2 = \text{Var}(X)$:

For a uniform distribution on $[a, b]$, the variance is:

$$\text{Var}(X) = \frac{(b-a)^2}{12} = \frac{(10-0)^2}{12} = \frac{100}{12} = \frac{25}{3}$$

Thus:

$$\boxed{\sigma_X^2 = \frac{25}{3}}$$

4. Calculate $\sigma_Z^2 = \text{Var}(Z)$:

Since X and Y are independent:

$$\text{Var}(Z) = \text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

We have $\text{Var}(X) = \frac{25}{3}$, and:

$$\text{Var}(Y) = \frac{(8-2)^2}{12} = \frac{36}{12} = 3$$

Thus:

$$\text{Var}(Z) = \frac{25}{3} + 3 = \frac{25}{3} + \frac{9}{3} = \frac{34}{3}$$

$$\sigma_Z^2 = \frac{34}{3}$$

5. Calculate $\text{Cov}(X, Z)$:

Recall that $\text{Cov}(X, Z) = E[XZ] - E[X]E[Z]$.

First, compute $E[XZ] = E[X(X+Y)] = E[X^2 + XY] = E[X^2] + E[XY]$.

Since X and Y are independent:

$$E[XY] = E[X]E[Y] = 5 \times 5 = 25$$

For a uniform distribution on $[a, b]$, $E[X^2] = \frac{a^2 + ab + b^2}{3}$:

$$E[X^2] = \frac{0^2 + 0 \cdot 10 + 10^2}{3} = \frac{0 + 0 + 100}{3} = \frac{100}{3}$$

Thus:

$$E[XZ] = E[X^2] + E[XY] = \frac{100}{3} + 25 = \frac{100}{3} + \frac{75}{3} = \frac{175}{3}$$

Now compute the covariance:

$$\text{Cov}(X, Z) = E[XZ] - E[X]E[Z] = \frac{175}{3} - (5 \times 10) = \frac{175}{3} - 50 = \frac{175}{3} - \frac{150}{3} = \frac{25}{3}$$

Alternatively, using properties of covariance:

$$\text{Cov}(X, Z) = \text{Cov}(X, X+Y) = \text{Cov}(X, X) + \text{Cov}(X, Y) = \text{Var}(X) + 0 = \frac{25}{3}$$

since $\text{Cov}(X, Y) = 0$ due to independence.

Thus:

$$\text{Cov}(X, Z) = \frac{25}{3}$$

Summary:

$$\mu_X = E[X] = 5$$

$$\mu_Z = E[Z] = 10$$

$$\sigma_X^2 = \text{Var}(X) = \frac{25}{3}$$

$$\sigma_Z^2 = \text{Var}(Z) = \frac{34}{3}$$

$$\text{Cov}(X, Z) = \frac{25}{3}$$

2. (20 points)

- (a) (10 points) Suppose that A is non-singular and (λ, x) is an eigenpair of A , show that (λ^{-1}, x) is an eigenpair of A^{-1} .

Proof. Since A is non-singular, $\lambda \neq 0$ (because if $\lambda = 0$, then $Ax = 0$ with $x \neq 0$, which would imply A is singular).

Given $Ax = \lambda x$, multiply both sides by A^{-1} on the left:

$$A^{-1}Ax = A^{-1}(\lambda x)$$

$$Ix = \lambda A^{-1}x$$

$$x = \lambda A^{-1}x$$

Since $\lambda \neq 0$, divide both sides by λ :

$$A^{-1}x = \frac{1}{\lambda}x$$

Thus, x is an eigenvector of A^{-1} with eigenvalue λ^{-1} , so (λ^{-1}, x) is an eigenpair of A^{-1} . $\square$

- (b) (10 points) For any scalar α that is not an eigenvalue of A , show that x is an eigenvector of A if and only if x is an eigenvector of $(A - \alpha I)^{-1}$.

Proof. ($\Rightarrow$) Suppose x is an eigenvector of A , so $Ax = \lambda x$ for some λ .

Consider $(A - \alpha I)x = Ax - \alpha x = \lambda x - \alpha x = (\lambda - \alpha)x$.

Since α is not an eigenvalue of A , we have $\lambda - \alpha \neq 0$ (otherwise $\alpha = \lambda$, making α an eigenvalue).

Multiply both sides by $(A - \alpha I)^{-1}$:

$$x = (\lambda - \alpha)(A - \alpha I)^{-1}x$$

Rearranging:

$$(A - \alpha I)^{-1}x = \frac{1}{\lambda - \alpha}x$$

So x is an eigenvector of $(A - \alpha I)^{-1}$ with eigenvalue $\frac{1}{\lambda - \alpha}$.

($\Leftarrow$) Suppose x is an eigenvector of $(A - \alpha I)^{-1}$, so $(A - \alpha I)^{-1}x = \mu x$ for some $\mu \neq 0$ (since $(A - \alpha I)^{-1}$ is invertible).

Multiply both sides by $(A - \alpha I)$:

$$x = \mu(A - \alpha I)x$$

Rearranging:

$$(A - \alpha I)x = \frac{1}{\mu}x$$

$$Ax - \alpha x = \frac{1}{\mu}x$$

$$Ax = \left(\alpha + \frac{1}{\mu}\right)x$$

Thus x is an eigenvector of A with eigenvalue $\alpha + \frac{1}{\mu}$. □

3. (20 points)

A random variable has probability density function

$$f(x; \theta) = (\theta + 1)x^\theta, \quad 0 < x < 1, \quad \theta \in \mathbb{R}.$$

Given samples $X_1, \dots, X_n$, derive the maximum likelihood estimator for the parameter θ .

Proof. The likelihood function is:

$$L(\theta) = \prod_{i=1}^n f(x_i; \theta) = \prod_{i=1}^n (\theta + 1)x_i^\theta = (\theta + 1)^n \prod_{i=1}^n x_i^\theta$$

The log-likelihood function is:

$$\ell(\theta) = \ln L(\theta) = n \ln(\theta + 1) + \theta \sum_{i=1}^n \ln x_i$$

Take derivative with respect to θ :

$$\frac{d\ell}{d\theta} = \frac{n}{\theta + 1} + \sum_{i=1}^n \ln x_i$$

Set derivative equal to zero to find critical points:

$$\frac{n}{\theta + 1} + \sum_{i=1}^n \ln x_i = 0$$

$$\frac{n}{\theta + 1} = - \sum_{i=1}^n \ln x_i$$

$$\theta + 1 = - \frac{n}{\sum_{i=1}^n \ln x_i}$$

$$\theta = -1 - \frac{n}{\sum_{i=1}^n \ln x_i}$$

Check second derivative to ensure maximum:

$$\frac{d^2\ell}{d\theta^2} = - \frac{n}{(\theta + 1)^2}$$

Since $n > 0$, the second derivative is negative for all $\theta \neq -1$, confirming we have a maximum.

Note: We require $\theta + 1 > 0$ for the PDF to be valid (needs to be positive). This imposes the constraint:

$$-1 - \frac{n}{\sum_{i=1}^n \ln x_i} > -1 \Rightarrow -\frac{n}{\sum_{i=1}^n \ln x_i} > 0$$

Since $n > 0$, this requires $\sum_{i=1}^n \ln x_i < 0$, which is true since $\ln x_i < 0$ for $x_i \in (0, 1)$.

Therefore, the maximum likelihood estimator is:

$$\hat{\theta}_{\text{MLE}} = -1 - \frac{n}{\sum_{i=1}^n \ln X_i}$$

Alternatively, we can write this as:

$$\hat{\theta}_{\text{MLE}} = -1 - \frac{1}{\frac{1}{n} \sum_{i=1}^n \ln X_i} = -1 - \frac{1}{\ln(\bar{X}_G)}$$

where $\bar{X}_G = (\prod_{i=1}^n X_i)^{1/n}$ is the geometric mean, since $\frac{1}{n} \sum_{i=1}^n \ln X_i = \ln(\bar{X}_G)$. $\square$

4. (20 points)

Given $p(x, y) = \frac{1}{25} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 2 & 1 & 2 & 0 & 0 \\ 2 & 0 & 1 & 1 & 1 \\ 0 & 3 & 0 & 2 & 0 \\ 0 & 0 & 1 & 1 & 3 \end{pmatrix},$

Marginals:

$$p_X(x) = \frac{1}{5}, \quad p_Y(y) = \frac{1}{5} \quad \forall x, y.$$

Entropies:

$$H(X) = \log_2 5, \quad H(Y) = \log_2 5.$$

Joint entropy: Let $n_{xy} \in \{0, 1, 2, 3\}$ be the integer entries. Counts: 8 zeros, 11 ones, 4 twos, 2 threes.

$$H(X, Y) = - \sum_{x, y} p(x, y) \log_2 p(x, y) = \log_2 25 - \frac{8}{25} - \frac{6}{25} \log_2 3.$$

Conditional entropies:

$$H(X|Y) = H(X, Y) - H(Y) = \log_2 5 - \frac{8}{25} - \frac{6}{25} \log_2 3,$$

$$H(Y|X) = \log_2 5 - \frac{8}{25} - \frac{6}{25} \log_2 3.$$

Mutual information:

$$I(X, Y) = H(X) + H(Y) - H(X, Y) = \frac{8}{25} + \frac{6}{25} \log_2 3.$$

Numerical values (approx.):

$$H(X) \approx 2.3219, \quad H(Y) \approx 2.3219, \quad H(X|Y) \approx 1.6215,$$

$$H(Y|X) \approx 1.6215, \quad I(X, Y) \approx 0.7004.$$

5. (10 points)

(i) Let X_n be number of umbrellas at Jim's current location at start of trip n . The weather each trip is independent with rain probability p . The next state X_{n+1} depends only on X_n and the rain status of trip n :

If $X_n = i \geq 1$ and it rains (prob p), Jim takes one umbrella: $X_{n+1} = 5 - i$. If $X_n = i \geq 1$ and no rain (prob $1 - p$), $X_{n+1} = 4 - i$. If $X_n = 0$ and rains, he gets wet but still $X_{n+1} = 4$; if no rain, $X_{n+1} = 4$ as well. Thus transitions are memoryless, so $\{X_n\}$ is a Markov chain on state space $\{0, 1, 2, 3, 4\}$ with transition matrix:

$$P = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1-p & p \\ 0 & 0 & 1-p & p & 0 \\ 0 & 1-p & p & 0 & 0 \\ 1-p & p & 0 & 0 & 0 \end{pmatrix}.$$

(ii) Initially $X_1 = 2$ (at home morning day 1).

Trip 1 (morning): If rain ($p = 0.2$) he takes an umbrella; if not (0.8) takes none. In either case he does not get wet because $2 \geq 1$.

Trip 2 (evening): After morning, X_2 is either 3 (if morning rain) or 2 (if morning no rain). In either case, if evening rain he has an umbrella to take. Thus probability he gets wet during day 1 is 0.

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6. (10 points)

We have

$$D_n^{(\alpha, \beta)}(1) = 1 - \left(\frac{\log \alpha}{\log n} \right)^\beta,$$

and for $x = 2, \dots, n+1$, $D_n^{(\alpha, \beta)}(x) = \frac{1}{n} \left(\frac{\log \alpha}{\log n} \right)^\beta$, otherwise 0.

For $\nu = (1, 0, 0, \dots)$,

$$D_{\text{KL}}(\nu \| D_n^{(\alpha, \beta)}) = \sum_x \nu(x) \log \frac{\nu(x)}{D_n^{(\alpha, \beta)}(x)} = -\log \left[1 - \left(\frac{\log \alpha}{\log n} \right)^\beta \right].$$

Let $\epsilon_n = \left(\frac{\log \alpha}{\log n}\right)^\beta$. As $n \rightarrow \infty$, $\log n \rightarrow \infty$, so $\epsilon_n \rightarrow 0$. Using $-\log(1 - \epsilon_n) \sim \epsilon_n \rightarrow 0$, we get

$$\lim_{n \rightarrow \infty} D_{\text{KL}}(\nu \| D_n^{(\alpha, \beta)}) = 0.$$

$$\boxed{\lim_{n \rightarrow \infty} D_{\text{KL}}(\nu \| D_n^{(\alpha, \beta)}) = 0}$$